The Local Economic Impact of Data Centers

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Abstract

A hyperscale data center costs over a billion dollars but employs only dozens of people. We ask whether these investments generate the local agglomeration spillovers that large plant openings have historically produced. Using satellite nighttime lights and within-site ring comparisons, we find a large but sharply localized capital footprint that fades within five kilometers, and no spillovers. Hiring, wages, supplier clustering, firm entry, and downstream industry growth are all absent, in both the ring data and realized county employment. The local incidence falls on land, as residential rents rise modestly without net in-migration. Data centers are agglomeration-scale capital without agglomeration.

Keywords: data centers; place-based investment; local labor markets; nighttime lights; agglomeration.

JEL Codes: R11, R23, O33, H71.

1 Introduction

The physical infrastructure of digital services has become a major component of US capital investment. Data centers consumed roughly four percent of US electricity in 2023, and the rise of artificial intelligence has sharply accelerated new construction. A single hyperscale campus typically costs five hundred million to two billion dollars and draws one hundred to five hundred megawatts of continuous power, yet employs only thirty to fifty people on site. At least thirty-five states now offer tax incentives aimed specifically at data centers, with cumulative subsidies approaching twenty billion dollars. Those subsidies are justified by the promise of local economic gains. This paper asks whether the gains materialize.

The natural benchmark is the canonical result on large plant openings. Greenstone et al. [2010] show that when a county wins a big new plant, the productivity of incumbent establishments rises by about twelve percent over the following five years. The gain operates through three agglomeration channels, namely a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. This result is the empirical backbone of the case for place-based subsidies.

A data center is agglomeration-scale in dollars but structurally unlike the plants that produced those spillovers. It employs dozens of people, not thousands, so it deepens no local labor pool. Its main inputs are electricity and fiber, purchased through national markets rather than from local suppliers. Its output is compute, delivered over the cloud to customers who may not be nearby. Each of the three channels requires either local labor or local input-output linkages, and a data center supplies neither. Whether the canonical agglomeration result extends to the signature capital investment of the AI era is therefore an open question, and a consequential one given the subsidies at stake.

We answer it with a research design built for the selection problem. Operators screen sites on cheap power, grid capacity, large parcels of low-cost land, and fiber, so a county that hosts a data center differs from one that does not in ways that also move the local economy. We document that this selection cannot be easily instrumented away: most predetermined features that predict hyperscale siting are correlated with local industrial character, and the features that are clean do not predict where these facilities go (Section 3). We therefore use two designs that do not require an instrument. The first compares rings around the same facility, restricting attention to isolated sites and dating each event to the start of construction, to recover the capital footprint in satellite nighttime lights. The second compares counties that gain a data center to counties that do not, in the realized employment data the prior literature uses, to recover the industrial composition of any employment effect. A comparison to announced-then-cancelled facilities serves as a falsification.

We first establish that the capital footprint is real, large, and local. Nighttime lights jump at the start of construction within the first kilometer of a hyperscale site and decay to near zero by five kilometers. Daytime imagery confirms the timing: vegetation at the parcel is cleared exactly when the lights turn on. We then turn to evidence for agglomeration. We test each of the three mechanisms at the facility. Hiring in the surrounding rings does not rise on net, local wages do not rise, and workplace employment shows no jump at opening. The sectors that build and equip a data center do not cluster near the site. The industries that use data centers do not subsequently form a local cluster, and new firm formation shows no jump in business-birth records. We confirm this in the realized county employment data, where the positive effect is confined to the data-center sector itself and its network, with no growth in any downstream or compute-using industry. Colocation data centers, which add capital but require little new construction, serve as a within-design placebo, showing no capital footprint and no spillover through any channel.

What local incidence we find falls on land, not labor. Residential rents rise modestly in the facility's catchment, with no multifamily permitting response, so the increase capitalizes into the existing housing stock rather than reflecting a demand boom. Net migration is near zero, because gross in-migration rises but is offset by comparable out-migration. Foot traffic falls in the inner rings of colocation facilities, where industrial use displaces consumer-facing activity, and commercial spending shows no robust response.

Taken together, the results describe capital without agglomeration. Data centers have recently been studied at the county level [Bahar and Wright, 2026, Alvarez et al., 2026]; our within-site design adds the sub-county capital footprint, direct tests of each agglomeration channel, and the incidence on land. Our contribution is threefold. First, we provide a sub-county test of whether agglomeration spillovers extend to the capital frontier of the AI era, and we find a clean boundary: they do not. Second, we identify the mechanism, that each agglomeration channel requires local labor or local input-output linkages that compute infrastructure does not create. Third, we trace the incidence to land and capital rather than labor, which bears on the design of the subsidies now flowing to these facilities. More broadly, the result marks a limit on the local-multiplier logic of industrial policy as investment shifts from labor-intensive plants toward labor-light digital capital.

The remainder of the paper proceeds as follows. Section 2 describes the data. Section 3 sets out the identification strategy. Section 4 documents the capital footprint. Section 5 tests the agglomeration channels and the county employment composition. Section 6 traces the local incidence to land. Section 7 interprets the findings. Section 8 concludes. Appendices report the instrument search, the geography of compute, and supplementary results.

2 Data

Data center registry. We build a master registry of US data centers from four sources. The base layer is the IM3 Open Source Data Center Atlas (Pacific Northwest National Laboratory), which identifies 1,242 facilities from OpenStreetMap footprints tagged as data centers, with coordinates and, where available, square footage and operator. We augment it with operational-date information from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, and trade press, and with four research batches targeting state announcements, specialty wholesale operators, AI-wave operators, and REIT filings. The final registry contains 1,494 facilities, of which 341 are hyperscale-tier and 339 are hyperscale-tier with operational dates inside the 2012–2024 VIIRS window. We assign hyperscale tier by operator identity (Amazon, Microsoft, Google, Meta, Apple, Oracle, plus hand-curated large specialty and AI-wave campuses); we do not impose a megawatt threshold, since capacity is not consistently observed. We also assemble a cohort of 54 announced-then-cancelled hyperscale projects for the falsification, and recover a construction-start year for each built facility from satellite-detected land clearing (a fall in vegetation cover paired with a rise in built-up surface).

Nighttime lights. We extract monthly mean radiance from the VIIRS Day/Night Band Monthly Composite for April 2012 through December 2024, in concentric rings of 0–1, 1–2, 2–5, 5–10, 10–25, and 25–50 kilometers around each facility, aggregated to annual means. We classify each facility as isolated if no other data center within two kilometers opened earlier, and use the isolated, construction-dated sample for the headline capital-footprint estimate.

Local labor, housing, and spending panels. We measure local labor demand from the LinkUp Job Records panel [LinkUp, 2025] (postings by employer and geocoded location, by ring and quarter), residential rents from the RentHub Rental Data panel [RentHub, 2022] (median list price by ring and month), consumer spending from the SafeGraph Spend Patterns panel [SafeGraph, 2022], and foot traffic from the Advan Weekly Patterns panel [Advan Research, 2025], all distributed by Dewey Data. Each enters the within-site ring design dated to operations, with state-by-month fixed effects.

Realized county employment. For the county-level composition analysis we use industry employment and establishment counts from the Quarterly Census of Employment and Wages, which align with the standard County Business Patterns employment and establishment measures. We focus on data processing (NAICS 518), telecommunications (517), web and information services (519), IT consulting (5415), and professional, scientific, and technical services (541), with manufacturing (31–33) and population-serving sectors as placebos.

Instrument-search inputs. The identification audit and instrument search draw on predetermined county features: 345 kV transmission-substation density, the InterTubes long-haul fiber network [Durairajan et al., 2015], the 1980 county share of urban college population (NHGIS), station-level climate normals (CustomWeather), and the EIA-860 generation fleet. None yields a usable sub-county siting instrument (Appendix A).

3 Identification

Data centers do not locate at random. Operators screen on cheap power, available grid capacity, large parcels of low-cost land, fiber proximity, and tax policy, so a county that hosts a data center differs from one that does not in ways that also move the local economy. A cross-county comparison would confound the data center with the conditions that attracted it. We document this selection directly: we tried to instrument data center siting with every plausibly exogenous predetermined feature, and each instrument failed, for reasons we report in Appendix A. The features that predict siting (power, industrial land, fiber) are themselves correlated with local industrial character, and the features that are clean (climate, historical human capital) do not predict where hyperscale facilities go. There is no clean instrument for hyperscale siting.

We therefore identify the local effect of a data center from two designs that do not require an instrument. The first compares rings around the same facility to recover the capital footprint. The second compares counties that host a data center to counties that do not, in the realized employment data the prior literature uses, to recover the sectoral composition of any employment effect. A cancelled-facility comparison serves as a falsification. The designs answer different questions and rest on different assumptions, and they agree.

3.1 The capital footprint: within-site rings, isolated sites, dated to construction

The capital footprint is the brightness a facility adds to its immediate surroundings, which we measure with satellite nighttime lights. The within-site specification compares an inner ring of a facility to the same facility's twenty-five to fifty kilometer outer ring,

$$\log(1 + \mathrm{NTL}_{i,r,t}) = \sum_k \beta_k \, \mathbf{1}[t - t_i^c = k] \times \mathrm{Near}_{i,r} + \alpha_{i,r} + \gamma_{r,t} + \varepsilon_{i,r,t}, \tag{1}$$

where i indexes the facility, r the ring, t the year, t_i^c the facility's construction-start year, $\mathrm{Near}_{i,r}$ marks the inner ring, $\alpha_{i,r}$ is a facility-by-ring fixed effect, and $\gamma_{r,t}$ is a ring-by-year fixed effect. The facility-by-ring fixed effect absorbs any time-invariant difference between a site's inner and outer ring, including the operator's siting preferences, the local industrial structure, and grid and fiber access. Identification comes from the change in the inner ring relative to the outer ring of the same site.

Two features of the design matter for credibility, and both correct a pre-trend present in a naive version of this specification.

First, we date the event to construction start, not to the operational opening. A hyperscale campus takes two to three years to build, so dating the event to the opening places the construction phase, when the site is physically lighting up, *before* the event window and reads it as a pre-trend. We recover the construction-start year for each facility from satellite-detected land clearing (a drop in vegetation cover paired with a rise in built-up surface). Dated to construction, the inner-ring brightness is flat before groundbreaking and jumps at it. Daytime imagery confirms the timing directly: vegetation cover at the parcel is flat until the construction year and falls sharply from it, so the event date marks a real physical change rather than a measurement choice.

Second, we restrict the headline to isolated sites, meaning facilities with no other data center within two kilometers that opened earlier. A new facility built beside an existing one inherits the neighbor's brightness in its inner ring before it breaks ground, which is the source of the remaining pre-trend. Splitting the sample makes this explicit. Among sites with an earlier neighbor within two kilometers, the inner-ring brightness rises fifty percent in the years before construction, the clustering artifact. Among isolated sites, where the inner ring is genuinely just the future parcel, the pre-construction excess is seven percent. The isolated-site, construction-dated event study has a flat pre-trend (a joint test of the pre-construction coefficients does not reject, $p = 0.21$) and a jump of thirty-eight percent in the construction year that builds to a large footprint over the following years. The jump survives an explicit control for a site-specific linear pre-trend, because the pre-trend, estimated from the pre-construction years, is essentially flat.

This design recovers the physical capital footprint. It does not, on its own, measure spillovers to the surrounding economy, which the remaining designs address.

3.2 Demand-side outcomes: the same ring design, dated to operations

The same within-site ring comparison in equation (1) identifies the demand-side outcomes that the incidence analysis uses: residential rents, consumer spending, and foot traffic. These respond when a facility begins operating rather than when it breaks ground, so we date them to the operational opening rather than to construction. The rent, spending, and foot-traffic panels are monthly and span the 2020 to 2023 period, so we replace the ring-by-year fixed effect with a state-by-month fixed effect, which absorbs the work-from-home and urban-recovery shocks that moved rents and visits sharply, and differently, across metropolitan areas in those years. We apply the same pre-period trend check, and, where the cancelled-facility panel allows, the same falsification: the rent response is present at built sites and absent at cancelled ones. The rent, spending, and foot-traffic estimates enter the incidence analysis in Section 6, where residential rents rise modestly at built sites, foot traffic falls in the inner rings as industrial land use displaces consumer-facing activity, and consumer spending shows no robust response.

3.3 The employment composition: counties that host versus counties that do not

Whether a data center raises local employment, and in which industries, is a question about the surrounding economy rather than the parcel, and it is naturally studied at the county level. We answer it in standard county employment data. We compare counties that gain a hyperscale data center to counties that do not, in an event study on County Business Patterns and Quarterly Census of Employment and Wages employment by industry,

$$\log(1 + \mathrm{Emp}^s_{c,t}) = \sum_k \theta^s_k \, \mathbf{1}[t - t^o_c = k] \times \mathrm{DC}_c + \mu_c + \lambda_t + u_{c,t}, \tag{2}$$

for industry s, county c, and the county's first data-center year t^o_c, with county and year fixed effects and never-treated counties as the comparison group. This design does not isolate the parcel, and it inherits the usual selection concern that data-center counties may trend differently. We read it through the sectoral composition of the effect, which is what distinguishes a direct effect from an agglomeration spillover, and we corroborate the contested sector with a staggered-robust estimator (Section 5.6).

The composition is informative on its own terms. Realized employment rises in the data-center sector itself (data processing, NAICS 518) and its network complement (telecommunications, NAICS 517), and it does not rise in any downstream or compute-using sector (web and information services, information-technology consulting, professional and scientific services). The same pattern holds for hyperscale counties, colocation counties, and the full set of data-center counties. The positive employment a county-level design records is the facilities and their direct network, not a spillover to surrounding firms.

3.4 A falsification: announced-then-cancelled facilities

A facility that was announced and sited but never built isolates the selection margin from the treatment. Cancelled and built sites were both chosen for the same grid and land features, so a difference between them after the scheduled opening reflects the data center rather than the siting. We assemble fifty-four announced- then-cancelled hyperscale projects and re-run the ring design on them. The capital footprint that built sites show is absent at cancelled sites, which confirms that the footprint is specific to a facility actually being built.

We treat the cancelled comparison as a falsification rather than the primary estimate, for two reasons. The cancelled sample is small, and most cancellations were driven by local opposition, so cancelled sites sit in systematically less development-friendly places and are not an exchangeable control. The design confirms the direction of the within-site result without carrying the identification.

3.5 Robustness

The capital-footprint estimate is robust to the usual checks on a ring design. Dropping each facility in turn leaves the estimate in a narrow band, so no single site drives it, and a donut specification that uses the one-to-two kilometer ring as the inner contrast, omitting the campus pixel, returns the same spatial decay. Because most hyperscale sites sit near another, we cluster standard errors on the campus rather than the facility, which widens the interval but leaves the effect significant. Most important for the construction-dating, we re-date the facilities from Federal Aviation Administration (FAA) obstruction-evaluation filings, a regulatory timestamp recorded years before any radiance response and joined to a facility only by location, with no satellite input. The construction-dated capital footprint is essentially unchanged under these independent dates, which establishes that it is not an artifact of nighttime-lights-derived event timing. For the rent result in Section 6, a synthetic-control estimator against population-matched non-data-center donors nets out the 2020 to 2023 densification that a within-site comparison would otherwise attribute to the facility, and places the data-center-specific rent rise in the two to five percent range, below the raw within-site estimate.

3.6 What does not identify the effect

For completeness, Appendix A reports the full set of instruments we tried. Every predetermined predictor of siting either fails an exclusion test or has no within-county power, and a shift-share instrument built from those predictors and national demand growth is too weak to identify a sub-county effect and loads on population-serving sectors, so it cannot separate the data center from general local development. Where it does identify, it agrees with the two designs above: the effect is in the direct data-center sector, not downstream. There is no clean sub-county instrument for hyperscale siting, which is why we rely on the within-site and county designs.

4 The capital footprint

The capital footprint is the brightness a facility adds to its immediate surroundings. It is the one outcome that is unambiguously present, and we document it before asking whether it produces any spillover.

4.1 The headline: a jump at construction in the inner ring

Figure 1 reports the within-site event study of equation (1) on isolated hyperscale sites, dated to the construction-start year. Inner-ring nighttime lights are flat in the years before groundbreaking and jump by thirty-eight percent in the construction year, building to a large footprint over the following years as the campus is completed and brought online. A joint test of the pre-construction coefficients does not reject a flat pre-trend ($p = 0.21$), and the jump survives a control for a site-specific linear trend estimated from the pre-construction years, because that trend is essentially flat. The footprint is the physical capital lighting up: the parcel, its security and parking lighting, its substations and cooling plant, and its continuous overnight operation.

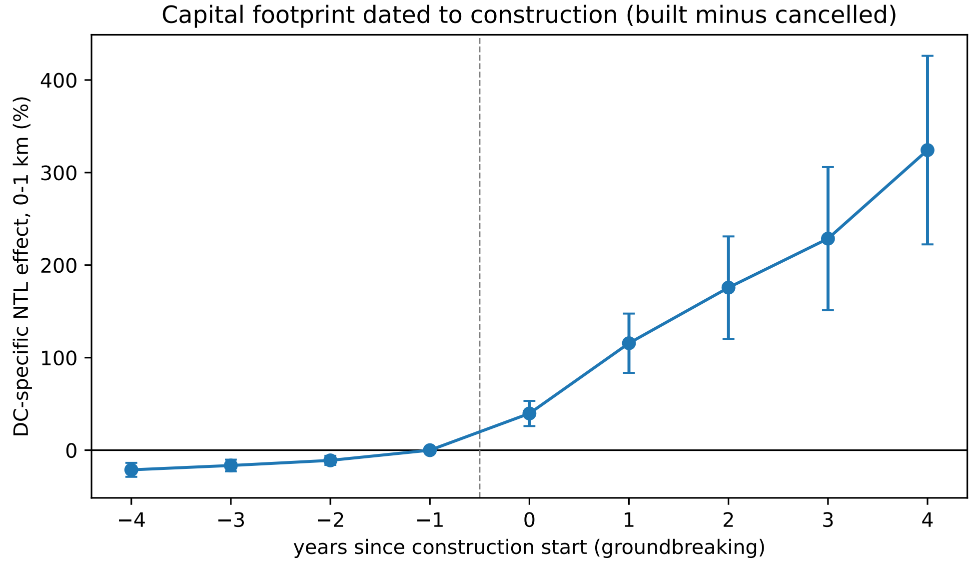

Figure 1: Capital footprint, dated to construction start (isolated sites)

Notes: Within-site event study of equation (1) on hyperscale facilities with a satellite-detected construction date and no other data center within 2 km that opened earlier (isolated sites). The plotted coefficients are the inner-ring (0–1 km) effect relative to the 25–50 km outer ring of the same site, by year since construction start, omitting the year before. Pre-construction coefficients are jointly indistinguishable from zero ($p = 0.21$). Standard errors clustered on facility.

Two choices make this estimate credible, and both correct a pre-trend that appears in a naive version of the design.

The first is the dating. A hyperscale campus takes two to three years to build, so dating the event to the operational opening places the construction phase, when the parcel is physically changing, before the event window and reads it as a pre-trend. We date each facility instead to the construction-start year, recovered from satellite-detected land clearing. Daytime imagery confirms the timing directly: at the parcel, vegetation cover is flat until the construction year and falls sharply from it, while built-up surface rises, so the event date marks a real physical change rather than a measurement choice.

The second is the sample. Roughly two in five hyperscale sites sit within two kilometers of an earlier data center, and such a site inherits its neighbor's brightness in its inner ring before it breaks ground. This clustering is the source of the remaining pre-trend. Splitting the sample makes it explicit. At sites with an earlier neighbor within two kilometers, inner-ring brightness rises about fifty percent in the years before construction. At isolated sites, where the inner ring is genuinely just the future parcel, the pre-construction excess is seven percent, and the event study is flat before groundbreaking.

4.2 Spatial decay

The footprint is sharply localized. Estimated ring by ring, the effect is largest at zero to one kilometer, falls by more than half at one to two kilometers, and is small and fading by two to five kilometers, with no detectable effect beyond five kilometers. A donut specification that drops the zero-to-one kilometer ring and uses one-to-two as the inner contrast returns the same decay, so the result is not an artifact of the campus pixel itself. The capital is large but it does not reach far.

4.3 Colocation as a within-design placebo

Colocation facilities add computing capacity inside existing or modest buildings and require little new construction. Run through the same design, they show no comparable capital footprint. The footprint is specific to the large physical builds that hyperscale campuses entail, which is what the rest of the paper takes as the treatment whose spillovers, or absence of them, we measure.

4.4 An outcome matrix

Table 1 summarizes every outcome at common coarse rings, hyperscale and colocation side by side, from the within-site ring difference-in-differences (DiD).

Table 1: Outcome matrix: hyperscale versus colocation

	Hyperscale			Colocation		
Outcome	0–5	5–10	10–25	0–5	5–10	10–25
Capital						
NTL (log radiance)	+6%***	null	null	+1%	−1%	−1%***
Displacement						
Foot traffic (visits)	+1%	+2%	−1%	−9%***	−8%**	−4%
Agglomeration						
Commercial spending	+8.5%	+10.4%	+9.9%	null	+4.8%	−3.0%
Median rent	+6.6%**	+9.4%***	+8.5%***	null	null	null
Postings (firm–DC)		+8%			+11%	
Workplace employment	null	null	null	+2%***	+2%**	null
High-earnings share	+0.20pp*	—	—	−0.47pp**	—	—
Agglomeration-channel tests (this paper)						
Supplier-sector postings	null	null	null	null	null	null
Advertised wage	null	null	null	null	null	null
New-firm births	null	—	—	—	—	—

Notes: Within-site ring difference-in-differences (equation 1) on the harmonized v8c backbone (1,113 data centers, 339 hyperscale), each outcome over a common 24-month post-opening window against the 25–50 km outer ring. NTL uses ring-by-year fixed effects; foot traffic, spending, and rent use state-by-month fixed effects, which absorb the 2020–2023 work-from-home shock; employment uses year fixed effects. Spending and rent are trend-break detrended; the data-center-specific rent rise is bracketed to 2 to 5 percent by a synthetic control (Section 6). Postings (firm–DC) is a pair event-study average over the same window with no ring structure. "null" denotes a coefficient not distinguishable from zero. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The matrix reports every outcome at the same coarse rings, including the null cells, so the ring-decay pattern is visible for each. The only outcome present at the inner ring is the capital footprint. The three direct agglomeration-channel tests in the bottom panel, namely supplier-sector postings, advertised wages, and new-firm births, are null at every ring.

Two caveats temper the matrix as a clean archetype comparison. Hyperscale and colocation facilities sit in structurally different places: the median 25 km catchment is 510 to 780 thousand people for hyperscale and 1.2 to 1.5 million for colocation, and hyperscale campuses sit a median 46 to 66 km from a top-50 metro centroid against 23 km for colocation. Some of what reads as an archetype difference may reflect the kind of place each prefers rather than the kind of facility it is. And most outcome windows overlap the 2020 to 2023 work-from-home period; the state-by-month fixed effects absorb it, but the colocation foot-traffic decline runs on a smooth secular slope with no inflection at opening, so part of that cell is not facility-driven.

5 A test of agglomeration

The capital footprint of Section 4 establishes that a hyperscale data center is a large, spatially concentrated investment. The question this section asks is whether that capital generates the local spillovers that large plant openings have historically produced. Greenstone et al. [2010] trace those spillovers to three channels: a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. We test each channel directly at the facility, using the same within-site ring design and the trend-break audit throughout. Colocation facilities, which add capital but require little new construction, serve as a within-design placebo. The result is uniform. None of the three channels operates, and the productivity outcome they would produce does not appear either.

5.1 Labor pooling

A deeper local labor pool is the first and most-studied agglomeration channel. If a data center deepened it, hiring and wages would rise in the surrounding rings. Neither does.

The ring-aggregated job-postings cell is null. Using the LinkUp panel under a Poisson pseudo-maximum-likelihood specification that handles the high frequency of zero counts, the change in postings near hyperscale facilities is statistically indistinguishable from zero at every ring, with a point estimate of -5.5% at 0–5 km ($p = 0.33$). A second, independent posting vendor agrees. The WageScape panel [WageScape, 2022] scrapes a different set of company sources and geocodes each posting to its city or ZIP centroid, and its count response near hyperscale facilities is also null at every coarse ring. Two vendors with non-overlapping coverage concur that hyperscale entry does not raise local labor demand.

The salary data reach a margin the counts cannot. The mean log advertised salary of nearby postings does not move after hyperscale entry at any ring. The estimate at 0–5 km is -1.3% ($p = 0.65$), measured over facility-by-ring-month cells that average several hundred salaried postings, so the null is well powered rather than imprecise. Workplace employment from the Longitudinal Employer-Household Dynamics (LODES) data tells the same story once the audit is applied. The raw within-site hyperscale cell is $+8\%$ at 0–5 km, but it does not survive trend-break detrending, collapsing to -0.3% with a highly significant pre-existing inner-versus-outer slope. The post-opening profile is the extrapolation of a trend, not a jump at opening.

Table 2: Job postings ring DiD: divergence from the nighttime-lights spillover

	DC-title	Non-DC-title	All postings
0–1 km	†	+5.9%*	+3.2%
1–2 km	†	+5.8%	+1.9%
2–5 km	-14.3%	+8.0%	+7.7%
Facilities in regression	298 hyperscale-operator postings DCs		

Notes: Poisson Pseudo-Maximum-Likelihood (PPML) estimates of within-site ring DiD on quarterly posting counts, aligned to v8c-curated operational dates. DC-title postings are those containing data-center-related keywords. Sample is hyperscale-operator-tagged postings (AWS, Microsoft, Google, Meta, Apple, Oracle, plus close variants). Pooled across archetypes. %lift reported as $(\exp(\hat{\beta}) - 1) \times 100$. † indicates a cell where the post-period count is essentially zero, producing PPML separation rather than an identified estimate. SEs clustered on DC. Stars: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

One labor margin is positive, and it is worth stating precisely because it does not overturn the null. A firm-DC pair event study on the LinkUp panel, identified off firm-by-facility fixed effects rather than ring-aggregated counts, finds that firms near a data center post more openings after it opens, with an early average treatment effect on the treated (ATT) of about +8% at hyperscale and +11% at colocation facilities over the first eight quarters. This is a recruiting-flow signal: known firms advertise more. It does not translate into the ring-level outcomes that a labor-pooling spillover would move. The ring-aggregated postings cell, the second vendor's counts, advertised wages, and detrended workplace employment are all null. Firms recruit toward the facility, but local labor demand and pay in the surrounding area do not rise.

While the level of this recruiting response is similar across archetypes, its composition differs. Table 3 reports the early ATT by curated technology-relevant NAICS subgroup. Hyperscale facilities pull a larger share of the response from compute-intensive subgroups, with software publishers up +24%, media streaming up +26%, and data-processing peers up +14%, against a flatter colocation distribution. Defining a core-compute basket as NAICS 5112, 512, 516, 517, and 518, the hyperscale early ATT is +23% on that basket against only +4% on other-labeled pairs at the same facilities, while the colocation gap runs the other way. This compute-intensive tilt is specific to hyperscale facilities, but it remains a composition feature of recruiting flow, not a realized local employment or wage gain.

Table 3: Postings early ATT by NAICS subgroup × archetype

NAICS subgroup	Hyperscale	Colocation
5112: Software publishers	+24%	+17%
512/516: Media streaming	+26%	+7%
517: Telecom carriers	+17%	+18%
518: Data processing / colocation	+14%	+8%
519: Web search / info services	+2%	+2%
5415: Tech consulting / IT services	+3%	+9%
5417: Pharma R&D	+9%	+16%
All firms (aggregate)	+8%	+11%

Notes: Early ATT is the firm-DC pair event-study average for quarters 0 through 8 post-opening (the common 24-month window), from `cf_069b`. Curated NAICS classification uses a hand-built technology-relevant grouping; row pair counts vary from 12–505 per cell (hyperscale) and 16–505 per cell (colocation).

5.2 Supplier linkages

The second channel is local supplier linkages. A large opening that bought inputs locally would pull supplier activity toward the site. We test this on the WageScape panel by isolating the sectors that build and equip a data center: construction, specialty trades such as electrical and mechanical contracting, electrical-equipment wholesale, architectural and engineering services, and machinery manufacturing. We aggregate postings in these sectors to facility-by-ring-month counts and run the within-site trend-break DiD on the composite and on each sector separately.

Supplier postings do not rise near hyperscale or colocation facilities. The hyperscale supplier composite is −1.9% at 0–5 km ($p = 0.87$), and every individual sector is null. Because supplier linkages can operate at a metropolitan rather than a hyperlocal scale, we also run a broad specification that compares the pooled 0–25 km near zone to a 50–150 km far control, which would catch a region-wide supplier rise that the within-site control might absorb. It is null as well, at +3.6% ($p = 0.73$) for hyperscale. Colocation is null under both designs. A pooled estimate across all facility types does show a positive composite jump of +14.4%, but it is driven entirely by two residual categories, unclassified facilities and a small set of cryptocurrency-mining sites, and vanishes for the hyperscale and colocation facilities that are the subject of the paper. The sectors that build a data center do not localize around it, consistent with build crews that travel and equipment that ships in from national markets.

5.3 Knowledge spillovers and firm entry

The third channel is knowledge spillovers among co-located firms. Two implications are testable. The industries that use data centers should cluster near them, and new firms should form in the catchment after a facility opens. Neither holds.

The downstream industry does not form a local cluster. Appendix B maps the firms that are the heaviest users of data-center capacity and shows that, while they sit somewhat closer to data centers than the average firm, the proximity reflects shared metropolitan location rather than anything the facility causes. Most large compute users are nowhere near a data center, the firms that are nearby pre-date the facility, and exurban data centers sit in compute deserts. The cloud decouples where compute is produced from where it is used, so the knowledge cluster that co-location would require never forms around the site.

New firm formation also shows no response at opening. We test it on true business births, which avoids the survivorship problem in registries of currently-active firms, using Secretary-of-State new-registration records for Virginia and Texas geocoded to the facility rings. The within-site trend-break jump in new registrations at the immediate 0–5 km ring is +0.5% ($p = 0.42$), a precise null. A placebo that moves the opening date thirty months earlier produces a null jump at every ring, which confirms that the design is not manufacturing the result from a pre-existing trend. The naive estimate at 0–5 km is negative, but it decomposes into no jump at opening plus a pre-existing decline, so the apparent drop in entry near data centers is a trend the facility does not change rather than an effect it causes. New firms do not enter the catchment when a data center opens.

5.4 Incumbent productivity

The three channels are the mechanisms; incumbent productivity is the outcome they produce, and the headline object in Greenstone et al. [2010], where incumbent establishments gain about twelve percent in total factor productivity. We cannot measure plant-level productivity directly without establishment output and capital data. We can bound it through its observable outlets. A productivity gain to incumbents shows up either as higher wages, since in a competitive labor market the wage equals the marginal revenue product of labor, or as faster growth and hiring at incumbent firms. Both are flat.

The wage outlet is the cleaner test, and it is the same well-powered null reported above. Advertised salaries in the catchment do not move after hyperscale entry, at any ring, on a panel that averages several hundred salaried postings per cell. Under the standard mapping from wages to marginal product, no local productivity gain reaches incumbent workers. The growth outlet points the same way but identifies less cleanly. A within-site event study on the hiring of firms that pre-date the facility shows no positive response at any horizon for hyperscale facilities, with every event-study coefficient statistically indistinguishable from zero. The pooled and colocation samples carry a pre-existing downward trend in inner-versus-outer hiring that the design cannot fully remove, so we read the incumbent-hiring evidence as corroborating rather than identifying. Both outlets rule out the large positive productivity response that the agglomeration benchmark would predict.

5.5 The pattern

Read together, the four tests describe a clean boundary. The labor pool does not deepen, suppliers do not localize, the downstream industry does not cluster, no new firms enter, and incumbent productivity, measured through wages, does not rise. The colocation arm, which adds capital but little construction, is null on every one of these margins as well, which is what a within-design placebo should show. A data center delivers agglomeration-scale capital without the agglomeration. The next section asks where the local value goes instead.

5.6 Realized county employment: direct, not downstream

The ring tests measure spillovers to firms near the facility, and they find none. A natural objection is that job postings are a thin measure: a data center's own jobs may be filled through corporate channels rather than posted to local boards, so a postings null could reflect measurement rather than the absence of jobs. We address this by turning to realized employment in standard county-level data, and by reading the effect through its industrial composition, which is what separates a direct effect from an agglomeration spillover.

We estimate the county event study of equation (2) on Quarterly Census of Employment and Wages employment, comparing counties that gain a hyperscale data center to counties that never do, for the data-center sector, its network complement, and the downstream industries that use compute. Table 4 reports the post-opening effect by industry and by facility type.

Table 4: County employment effect by industry and facility type

Industry (NAICS)	Hyperscale	Colocation	All data centers
Direct and network			
Data processing (518)	$+56^{***}$	$+37^{***}$	$+38^{***}$
Telecommunications (517)	$+43^{*}$	null	null
Downstream / compute-using			
Web and information services (519)	null	null	null
IT consulting (5415)	null	null	null
Professional, scientific, technical (541)	null	null	null

Notes: Post-opening effect (percent) from the county event study of equation (2), county and year fixed effects, never-treated counties as the comparison group, standard errors clustered on state. "null" denotes a coefficient not distinguishable from zero at the ten-percent level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The composition is unambiguous. Realized employment rises in the data-center sector itself and in telecommunications, the network complement that data centers carry with them. It does not rise in any downstream or compute-using industry, and this holds for hyperscale counties, colocation counties, and the full set of data-center counties. The positive employment a county-level design records is the facilities and their network, not a spillover to surrounding firms. The accompanying establishment counts confirm the reading: the rise in data-processing establishments is the data centers and colocation providers themselves, while establishment counts in the downstream industries are flat, so there is no firm entry to support an agglomeration interpretation.

This resolves the measurement objection cleanly, sector by sector. The postings null in the data-center sector is measurement, since the operators' own jobs do not appear on local boards while the realized employment in Table 4 does. The postings null in the downstream industries is real, since the realized county employment is null there as well. "No hiring spillover" is therefore a statement about the data, not an artifact of the postings source.

The web and information services sector (519) is the one downstream industry where a level-matching design can appear to show an effect. It does not survive a staggered-robust estimator. Under a Callaway–Sant'Anna design with not-yet-treated controls, the post-opening effect is small, statistically insignificant, and no larger than the sector's own pre-period leads, and its establishment count is flat throughout, so there is neither firm entry nor anticipatory co-location. The apparent signal is a pre-trend rather than an effect.

6 Incidence on land

The agglomeration channels are absent, but the local economy is not entirely unmoved. A modest amount of value does stay in the catchment. It accrues to land rather than to labor, and it looks like capitalization rather than a demand boom.

Residential rents rise near hyperscale facilities. The within-site detrended rent cells are +6.6% at 0–5 km, +9.4% at 5–10 km, and +8.5% at 10–25 km, with the peak in the broader residential catchment rather than immediately at the campus. Two cautions pull the headline number down. An independent synthetic-control estimator against the 3,000 non-DC tract donor pool returns smaller per-facility magnitudes, and a density-matched donor placebo shows that random non-DC sites of comparable density also exhibit rent growth of about +8.4% at 5–10 km during the 2019–2023 work-from-home window. Part of the within-site coefficient therefore reflects a general densified-area pattern rather than a data-center effect. We read the data-center-specific rent residual as +2 to +5% at the 5–10 km peak, with the synthetic control as the headline and the within-site estimate as an upper bound. Colocation rent is null at every ring and in every robustness specification.

The rent rise arrives without the supply or demand signatures of a local boom. A place-level Census Building Permits Survey check finds no multifamily permitting response in the same ring, with 5-or-more-unit permits up only +3.4% ($p = 0.87$). On the demand side, a county-to-county migration event study on IRS Statistics of Income (SOI) data shows in-migration rising to +6.3% and adjusted-gross-income inflow to +9.0% by year five after a hyperscale opening, with a clean colocation null. But out-migration rises by a comparable amount over the same window, so the net change in residents is near zero. The gross flows describe churn and a shift in composition toward higher-income in-migrants, not population growth. With no new supply and no net new residents, the rent increase is best read as capitalization into the existing housing stock.

Two further pieces fit this reading. Local public finances strengthen after a hyperscale opening, with larger school revenues and higher capital outlay and public-works spending, so part of the measured rent rise plausibly capitalizes local fiscal capacity and infrastructure rather than labor-market demand. And commercial spending rises weakly at 0–5 km, by about +8.5% on the cleaned point-of-interest panel, though the within-site standard error leaves it imprecise ($p = 0.50$); it passes a random-donor placebo but we do not treat it as a separately identified channel. The spending response is consistent in direction with the rent and migration evidence and equally consistent with capitalization rather than a new local consumer base.

Figure 2 reports the migration event study in full. The hyperscale in-migration rise survives a pairs cluster bootstrap (199 resamples of county fips), with in-migration of +6.3% at $p = 0.010$ and AGI inflow of +9.0% at $p < 0.005$ by year five, while the colocation in-migration cell stays null. The pre-period is short and the treated-county count modest, so we read the migration arm as corroborating the capitalization story rather than as a standalone identification.

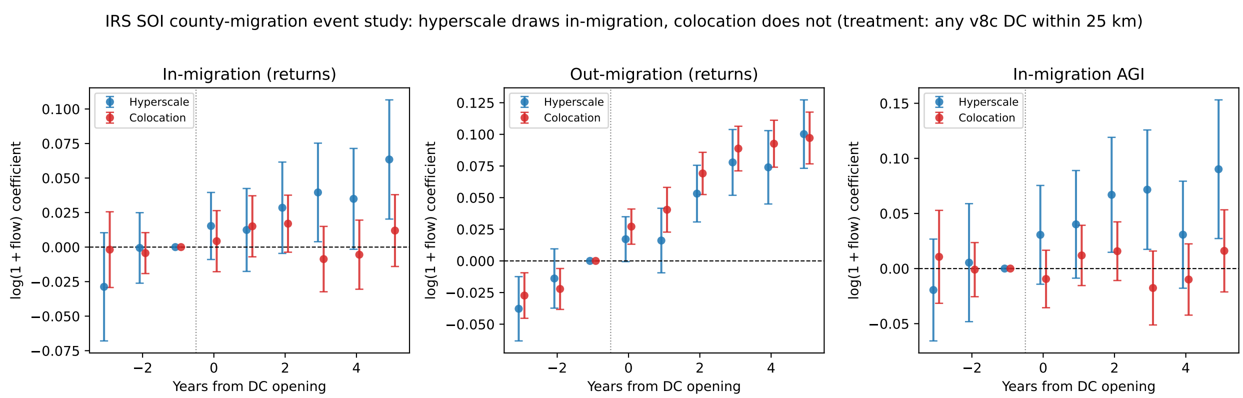

Figure 2: IRS SOI county-migration event study around DC openings. Hyperscale openings are followed by rising in-migration of returns and AGI; colocation openings are null on in-migration.

Notes: County-level event study of log return-inflow, log return-outflow, and log AGI-inflow around v8c data-center openings (`cf_240`). Treatment: county is treated at year $t^\star$ if any v8c high- or medium-confidence DC of the listed archetype sits within 25 km of the county centroid and opened in year $t^\star$. Hyperscale arm: 76 treated counties, 3,145 control counties; colocation arm: 191 treated counties, 3,030 control counties. Specification: $\log(1 + \text{flow})$ on event-time dummies d_e for $e \in \{-3, \ldots, +5\} \setminus \{-1\}$ with county and state × year fixed effects; CRV1 standard errors clustered on county fips. Reference period is $e = -1$. Inflow / outflow / AGI are the IRS "Total Migration-US" rows aggregating all US domestic flows into / out of each county-year. Markers offset ±0.08 on the x-axis for visibility. 95% confidence intervals shown.

The fiscal evidence reinforces the capitalization reading. Table 5 combines school-district and non-school local-government finance panels. Hyperscale openings are followed by larger local school revenues, higher school capital outlay per pupil, and higher county highway and public-works spending. Part of the measured rent rise is therefore consistent with households capitalizing local fiscal capacity and infrastructure improvements that arrive with hyperscale development.

Table 5: Public-goods capitalization checks

Outcome	Geography / sample	Hyperscale	Colocation
Local property-tax revenue	school district	+24.6%***	+8.9%***
Total local revenue	school district	+12.8%***	+3.8%*
Capital outlay per pupil	school district	+35.0%*	+4.3%
Property-tax revenue	place, balanced	+20.2%***	+10.6%***
Highway current + construction	county, balanced	+47.6%***	+10.9%
Broad public works current + construction	county, balanced	+18.7%**	+11.0%*
Public-building current + construction	county, balanced	+75.9%**	+27.2%

Notes: School-district estimates use NCES/Census F-33 finance files for fiscal years 2012–2023 (`cf_210`). Local-government estimates use Census Annual Survey/Census of Governments individual-unit files for fiscal years 2012–2023 (`cf_211`). All specifications use geography fixed effects and state-by-year fixed effects, with standard errors clustered by geography. The local-government rows report the balanced 2012–2023 panel. Public-works outcomes use the consistently available current plus construction margin (Census item families E** + F**); other-capital-outlay G** codes are excluded because they do not appear consistently in the 2022–2023 public-use files. Stars: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The incidence is therefore narrow and lands on owners. The surplus that stays local shows up in higher land and housing values and in stronger municipal balance sheets, captured by landowners and local governments, alongside the operator's own capitalized footprint in the nighttime-lights signal. Local workers and local firms, the parties an agglomeration spillover would benefit, are left roughly where they started.

7 Interpretation

The pattern is simple to organize. A large plant opening reaches the local economy through three agglomeration channels: a deeper shared labor pool, denser supplier linkages, and knowledge spillovers among nearby firms. Each channel requires either local labor or local input-output linkages, and a data center supplies neither. It employs dozens of workers, not thousands, so it deepens no labor pool. It buys electricity and fiber through national markets, so it localizes no suppliers. It ships compute over the cloud to distant customers, so it seeds no knowledge cluster. The channels are absent by construction, which is why a footprint this large does not propagate.

What remains is the capital itself and a narrow incidence on land. The footprint is the facility's own physical capital, large and confined to the inner kilometer. Its only local transmission runs through housing: a modest rent increase in the catchment that capitalizes into the existing stock. The transmission is small because the new residents are few, and it shows up as a price rather than as quantity because the housing stock is fixed within the horizon we observe. The relevant constraint is the size of the existing stock relative to the shock, not the long-run supply elasticity, which has no time to operate. This is why the percentage rent response is larger in small exurban housing markets, where hyperscale campuses tend to locate, than in dense metros.

The contrast between the two facility types follows the same logic. Hyperscale facilities are large greenfield builds, so they show the capital footprint and the rent response. Colocation facilities retrofit existing space and add capacity rather than construction, so they show neither, and they serve as a within-design placebo throughout. Because the agglomeration channels are absent by the structure of the technology rather than by happenstance, the conclusion is unlikely to reverse as the sector grows.

8 Conclusion

A hyperscale data center is as large a capital investment as the manufacturing plants that anchor the agglomeration literature, but our evidence shows it does not behave like one. The capital footprint is large and local, and it stops there. Nighttime lights jump at the start of construction within the first kilometer of a hyperscale site, restricting to isolated facilities and dating the event to groundbreaking so that the build is not mistaken for a pre-trend, and daytime imagery confirms the timing. None of the three channels through which a large plant raises local productivity operates. The local labor pool does not deepen, suppliers do not localize, the downstream compute industry does not cluster, and no new firms enter. In the realized county employment data, the gains are confined to the data centers and their direct network. Colocation facilities, which add capital without new construction, are null on every margin.

The only local outcome that moves with the capital is land. Residential rents rise by a few percent at built sites and are flat at cancelled ones, the rise arrives without new multifamily permitting and without net in-migration, and gross in-migration is offset by comparable out-migration, so the increase capitalizes into the existing housing stock. Colocation foot traffic falls in the inner rings, and commercial spending does not move. The local incidence is narrow. It falls on landowners and on municipal balance sheets, not on local workers or firms.

Three implications follow for place-based incentives. A jurisdiction that recruits a hyperscale facility should expect a large physical footprint and a modest capitalization into land values, but not the labor-market or supplier spillovers that justify subsidies aimed at conventional plants. The roughly twenty billion dollars committed to attracting these facilities is buying capital, not agglomeration. The incidence falls on owners of land and on local public finances rather than on workers, which matters for who gains and who pays when a subsidy is financed locally. And because the agglomeration channels are absent by construction rather than by happenstance, the result is unlikely to reverse as the sector grows: a data center deepens no local labor pool, buys its inputs through national markets, and ships its output over the cloud, so the local-multiplier logic that animates industrial policy does not extend to labor-light digital capital.

A The instrument search

A natural way to address the siting-selection problem is an instrument for data center location. We searched for one and did not find a clean one. This appendix documents the search, because the failure is informative: it is the reason we identify the effect from the within-site and county designs rather than from an instrument, and it characterizes how hyperscale siting actually works.

Predetermined siting instruments. We tested each plausibly exogenous, predetermined predictor of data-center siting, interacted where needed with the national capacity wave, and screened each against a manufacturing placebo (a valid instrument for data-center activity should not predict manufacturing employment). Table 6 summarizes the outcomes. Transmission-substation density predicts siting strongly but fails the placebo, because it is a proxy for industrial land. The 1980 urban college share, which works in the all-data-center county literature, carries the wrong sign for hyperscale, because hyperscale campuses go to low-college exurban land rather than to educated metros. Long-haul fiber proximity predicts siting only across metropolitan areas, not within a county, so it cannot identify a sub-county effect. Climate, measured by wet-bulb temperature and free-cooling days, does not predict siting at all, because the largest US hyperscale hub is hot and humid and operators engineer around climate to chase cheap power. Generation-capacity timing has no within-county predictive power. The pattern is systematic rather than bad luck: the features that predict hyperscale siting (power, industrial land, fiber) are the same features that are correlated with local industrial character, and the features that are clean (climate, historical human capital) do not predict where these facilities go. Hyperscale siting is cost-driven, and there is no clean predetermined instrument for it at sub-county scale.

Table 6: Predetermined siting instruments and why each fails

Instrument	First stage	Why it fails
345 kV substation density	strong	fails manufacturing placebo (industrial proxy)
1980 urban college share	moderate	wrong sign (hyperscale $\rightarrow$ low-college exurbs)
InterTubes fiber proximity	metro-scale	null within county
Climate (wet-bulb, free-cooling)	null	siting is cost-driven; the main hub is hot-humid
Generation-capacity timing	null	no within-county power

Notes: Each instrument is the predetermined county feature interacted with the national hyperscale capacity wave. "Fails placebo" means the instrument predicts manufacturing employment, violating the exclusion restriction.

A shift-share instrument is also weak. A shift-share design of the form used in concurrent county-level work [Alvarez et al., 2026], interacting the predetermined fiber and college exposures with national data-center demand growth, fares no better here, and the obstacle is the endogenous variable. Our public measures of data-center scale, namely capacity, square footage, and facility count, are lumpy and supply-driven, and a demand-side shift predicts a lumpy supply-side object poorly. The first stage with square footage as the endogenous reaches an F of only twenty to thirty, too weak to use.

Where it does identify, it agrees with the rest of the paper. The two-stage least-squares effect loads on the direct data-center sector and is null or negative in the downstream industries, the same composition the county event study recovers. The instrument also loads on population-serving sectors such as health, the general area-development confound that the within-site and cancelled designs avoid, which is why we treat it as a robustness check rather than a primary design.

B The geography of compute

A common rationale behind data center incentives is that the facility will anchor a local digital economy. If the cloud servers are here, the reasoning goes, the firms that use the cloud will follow. This section examines that premise directly by mapping where the heavy users of compute actually sit relative to data centers, and by asking whether their location reflects the data center or something the data center does not provide. We use the Veridion firmographic database [Veridion, 2026], which geolocates 6.5 million US establishments and classifies each by industry, employment, and founding year. We define compute users by primary industry: software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services form the compute-native core, with finance, scientific R&D, and semiconductors as a broader set.

B.1 Compute users cluster near data centers

Compute users are closer to data centers than the average firm. The median compute-native firm sits 11 kilometers from the nearest data center, against 20 kilometers for non-compute firms, and 69 percent of compute users fall within 25 kilometers of a data center, against 55 percent of other firms (Table 7). The gradient runs as one would expect from the technology. Cloud and hosting firms are nearest at a 6 kilometer median, software firms next at 9 kilometers, while telecommunications and streaming, whose footprints are more dispersed, sit farthest out.

This proximity is not an artifact of headquarters location. Measuring each firm's operating footprint by the geography of its job postings rather than its headquarters returns the same pattern at a slightly larger distance, because a firm's branch offices spread beyond its headquarters metro. Nor is it merely that both compute users and data centers crowd into a handful of tech metros. Removing the 45 largest metropolitan areas, compute users remain closer to the surviving data centers than non-compute firms are (a 75 versus 87 kilometer median, with software firms among the closest), so the association holds in second-tier and rural America as well.

Table 7: Distance from firms to the nearest data center

	Firms	Median km	$\leq$ 25 km	$\leq$ 50 km
Panel A: Headquarters location (Veridion)				
Non-compute firms	5,692,641	20.5	55%	70%
Compute users (core)	309,760	11.0	69%	81%
Cloud and hosting	26,188	6.1	75%	84%
Software	101,757	8.9	72%	84%
Computer systems design	154,934	11.7	68%	81%
Finance	193,540	15.9	62%	75%
Telecommunications	10,125	22.2	53%	66%
Panel B: Establishment footprint (job postings)				
Compute users (core)		15.1	63%	76%
Panel C: Outside the 45 largest metros				
Non-compute firms	2,603,195	86.8	20%	33%
Compute users (core)	103,356	75.3	30%	41%

Notes: Distance from each firm to the nearest hyperscale-clean data center, by haversine. Panel A places each firm at its Veridion headquarters coordinate. Panel B replaces the headquarters with the firm's job-posting locations (WageScape), weighted by the number of distinct hiring firms at each location, so it reflects the operating footprint rather than the headquarters. Panel C excludes all firms and data centers within 50 km of the 45 largest metropolitan and technology-hub centers and measures distance to the nearest remaining data center. Compute-native "core" sectors are software publishing, computing infrastructure and hosting, computer systems design, and web and streaming services.

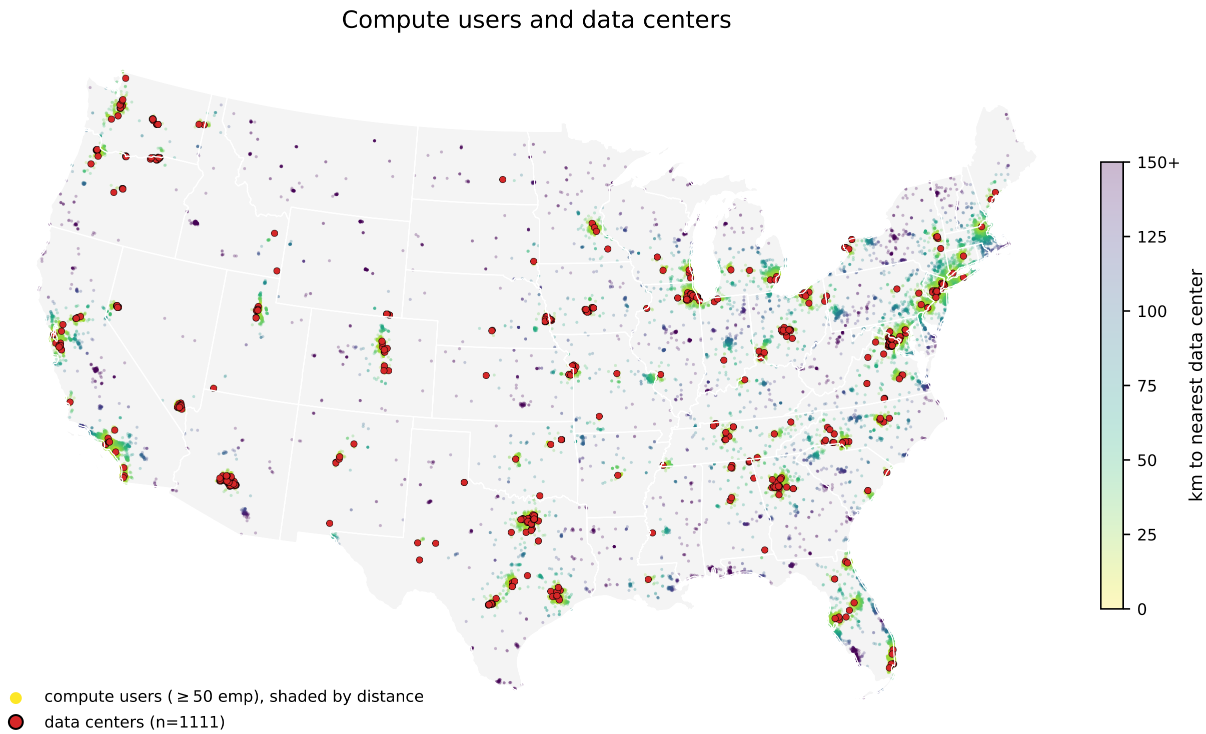

Figure 3: Compute-user firms and data centers across the continental United States. Points are compute-native firms with at least 50 employees, shaded by distance to the nearest data center, with data centers marked in red. Both concentrate in the same coastal and metropolitan corridors, while interior firms shade dark, far from any facility.

The two populations trace the same map (Figure 3), crowding into the coastal and metropolitan corridors and thinning together across the interior. The pull is strongest for the largest firms. When sorted by employment, the median distance to a data center falls monotonically from 13 kilometers for the smallest firms to 4 kilometers for those with more than ten thousand employees, and the dispersion collapses with it. The biggest compute users cluster tightly around data centers, and almost none are far from one. Only about 5 percent of compute firms with at least a thousand employees sit more than 50 kilometers from any data center. The handful that do are software companies that scaled in place in second-tier cities, such as Esri in Redlands, Jack Henry in rural Missouri, and Ansys outside Pittsburgh. Distance to compute is a constraint these firms evidently did not face, because the cloud let them consume capacity hundreds of kilometers away.

B.2 Two kinds of data center hub

The co-location is a metropolitan phenomenon, and seeing this matters for what follows. Grouping data centers into hubs reveals a sharp split. Metropolitan hubs are embedded in dense clusters of sector-matched compute users. The 239 data centers around Ashburn, Virginia sit among thousands of government IT contractors, with DXC, Leidos, Booz Allen, SAIC, and CACI all within 25 kilometers. The Bay Area hubs are ringed by Google, Broadcom, Synopsys, and ServiceNow. By contrast, the large exurban hyperscale clusters are isolated. The 50 data centers around Hillsboro, Oregon and the cluster near Quincy, Washington each have on the order of four compute firms within 25 kilometers. The same physical infrastructure produces a thick compute neighborhood in a tech metro and an empty one in the exurbs.

B.3 Why they co-locate, and why it is not the data center

Part of the explanation is shared infrastructure. Data centers and compute users both require access to the high-voltage grid, and both sit closer to it than ordinary firms. The median data center is 3.4 kilometers from a 230-kilovolt substation, the median compute user 5.0 kilometers, and the median other firm 6.4 kilometers. High-voltage substation density independently predicts where compute users locate. But the grid is only part of the story. The proximity between compute users and data centers survives controlling for substation access, so a common pull toward the power backbone, together with the fiber and talent that concentrate in metros, leaves the two in the same places, consistent with shared infrastructure rather than one drawing the other.

Three pieces of evidence indicate that the data center is not the cause. First, the compute users near data centers predate them. Among compute firms within 25 kilometers of a data center, at least 69 percent were founded before the nearest data center opened. Because the firm registry over-represents recently founded firms, the true share is higher. The local compute base existed first. Ashburn illustrates the direction of causation. The government technology ecosystem of Northern Virginia and its early internet exchange drew the data centers to the region, not the reverse. Second, the timing evidence in Section 3.5 shows that data center openings do not raise job postings in the surrounding rings, for compute-using industries or any other, once a smooth pre-existing trend is removed. The opening is not a turning point for local hiring. Third, and most directly, exurban data centers sit in compute deserts and stay there. The median metropolitan data center has roughly 2,700 compute firms within 25 kilometers, while the median exurban data center has 41 (Table 8). The facility does not move that number.

Table 8: Compute users near metropolitan versus exurban data centers

	Data centers	Median compute firms $\leq$ 25 km	Share with $<$ 5 nearby
Metropolitan data centers	824	2,702	0%
Exurban data centers	287	41	11%

Notes: Each data center is classified as metropolitan if it lies within 50 km of one of the 45 largest metropolitan or technology-hub centers, and exurban otherwise. The middle column reports the median count of compute-native firms (Veridion) within 25 km of the data center; the last column the share of data centers with fewer than five compute firms within 25 km. Separately, among compute firms within 25 km of any data center, 69 percent were founded before the nearest data center opened.

B.4 Implication for place-based policy

The policy reading is direct. A jurisdiction weighing a data center, especially an exurban one, should not expect compute users to arrive in its wake. Firms that consume cloud capacity locate near transmission, fiber, and skilled labor. These assets concentrate in metropolitan areas, and a single data center does not create them. The compute users already near data centers were mostly there first, and the data centers that open in compute-sparse places remain in compute-sparse places. What the facility delivers locally is the capital footprint and the rent pressure documented above, not a cluster of digital firms. The hope that the servers will bring the software is, on this evidence, misplaced.

C Supplementary results

C.1 Phase decomposition: construction versus operations

The matrix and the fine-ring decomposition treat the within-site nighttime-lights footprint as a single object that switches on at the operations date. We can split it further. Roughly one quarter of the hyperscale footprint accrues during the construction phase, before operations begin, and the remaining three quarters arrives when operations come online. This is the central reading of Table 9.

We anchor the construction-start date on three independent sources. The first is the sustained drop in the Sentinel-2 normalized difference vegetation index (NDVI) that marks land clearing on the facility footprint (`cf_097`). The second is the earliest FAA Obstruction Evaluation Form 7460-1 filing within 500 meters of the facility, restricted to construction-relevant structure types (cranes, transmission-line towers, buildings, water systems) and to filings within a plausible window around opening (`cf_310`). The third uses the same FAA filter but at a 1 kilometer radius for a larger sample at the cost of admitting some neighbor-project contamination. The three sources draw on different sensing modalities (optical satellite, regulatory filing) and different spatial radii, so convergent estimates are informative about the underlying object rather than about any one timestamp's measurement properties.

We replace the single post-opening indicator in the within-site DiD with two phase indicators. `post_construction` equals one for facility i in year-quarter t when t is on or after the construction-start date and before the operations date; `post_operations` equals one when t is on or after the operations date. We interact each with the inner-ring indicator and estimate the two interactions jointly on the standard near-versus-outer panel (0 to 5 km inner, 25 to 50 km outer). The construction-phase ATT is the coefficient on `near` $\times$ `post_construction`; the operations-incremental ATT is the coefficient on `near` $\times$ `post_operations`; the operations-total ATT is their sum, with the joint standard error from the variance-covariance matrix.

The three samples converge on point estimates although not on precision. On the NDVI-dated sample (`cf_300`, $n = 43$ hyperscale), the construction-phase ATT is +8.0% ($p = 0.015$) and the operations-total ATT is +29.5% ($p < 0.001$). On the FAA-dated 500 meter sample (`cf_300b`, $n = 64$), the construction-phase point estimate is similar at +8.7% but not statistically distinguishable from zero ($p = 0.14$), and the operations-total ATT is +33.3% ($p = 0.03$). On the FAA-dated 1 kilometer sample ($n = 119$), the construction-phase ATT is +6.2% ($p = 0.07$) and the operations-total ATT is +28.4% ($p = 0.001$). The point estimates lie in a tight band: construction-phase between 6 and 9 percent and operations-total between 28 and 33 percent. The implied construction share of the total is 0.22 to 0.27 across all three samples. The colocation rows in Panel B are statistically indistinguishable from zero on both phases in every sample. The decomposition therefore inherits the matrix's archetype contrast: hyperscale generates a sizable pre-operations footprint that colocation does not.

Two caveats temper the construction-phase point estimate. First, the FAA `receivedDate` is a permit filing date that precedes ground-breaking by roughly 3 to 12 months, so the FAA construction window includes some pre-construction calendar months in which no nighttime-lights change is expected. This biases the FAA construction-phase estimate downward relative to the bulldozers-to-operations true ATT and partially accounts for the larger standard errors in the FAA samples relative to NDVI. Second, identification of the phase decomposition leans on cross-facility variation in the construction-to-operations lead time. The interquartile range of lead times remains wide in all three samples (11 to 50 months at the FAA 1 kilometer anchor), which supports identification, but tight clusters of lead times around a single duration would not.

The decomposition has a substantive interpretation. The construction-phase NTL signal is consistent with on-site activity that begins before commercial operations: perimeter security lighting, substation and switchgear installation, cooling-tower assembly, fiber-conduit work, and the construction-crew lighting that accompanies a multi-year hyperscale build. The operations-phase increment captures the remaining lift after commercial illumination, server-floor load, and continuous twenty-four-hour cooling come online. Roughly three quarters of the total within-site footprint is operations; roughly one quarter is construction. The latter is a temporary economic-activity signal that ends when the facility opens; the former persists for as long as the facility operates.

Table 9: Phase decomposition of the hyperscale within-site nighttime-lights footprint

Timing source	n_{DC}	Construction-phase ATT (%)	Operations total ATT (%)	Construction share of total
Panel A: Hyperscale				
Sentinel-2 NDVI clearing	43	$+8.0^{**}$	$+29.5^{***}$	0.27
		(3.4)	(11.0)	
FAA OE permit, 500 m	64	+8.7	$+33.3^{**}$	0.26
		(6.0)	(15.4)	
FAA OE permit, 1 km	119	$+6.2^{*}$	$+28.4^{***}$	0.22
		(3.4)	(8.4)	
Panel B: Colocation				
Sentinel-2 NDVI clearing	21	−1.0	−1.4	—
		(2.3)	(4.3)	
FAA OE permit, 500 m	45	−5.3	−9.2	—
		(3.7)	(7.2)	
FAA OE permit, 1 km	120	−2.6	−3.6	—
		(2.1)	(3.9)	

Notes: Each row is a separate within-site ring difference-in-differences with two phase indicators replacing the single post-opening indicator: `post_construction` equals one between the construction-start date and operations, and `post_operations` equals one after operations begin. Coefficients are reported in percent ($100 \times (e^{\beta} - 1)$). Operations-total ATT is the sum of the construction-phase and incremental-operations coefficients, with standard errors computed from the joint variance-covariance matrix. Construction-start dates come from three independent sources: Sentinel-2 NDVI sustained vegetation drop (cf_097), and the earliest project-tied FAA OE Form 7460-1 filing within 500 m or 1 km of the facility (cf_310). Standard errors are clustered on dc_id. The 500 m FAA sample excludes ambiguous matches; the 1 km sample is the larger but noisier comparator. The downward bias on the construction-phase point estimate in the FAA samples reflects that the permit-filing date precedes ground-breaking by roughly 3 to 12 months. Construction share = construction-phase ATT divided by operations-total ATT. *, **, *** indicate $p < 0.10$, 0.05, 0.01.

C.2 New business formation: a contextual heterogeneity finding

A natural follow-up question is whether the A-channel commercial-activity lift around hyperscale openings is driven by *new* firms entering the catchment or by *existing* firms expanding. Neither the SafeGraph Spend panel (transactions at existing POIs) nor the LinkUp postings panel (recruiting at existing firms) speaks directly to firm entry. To address this, we assembled state-level business-registration panels for Virginia, Texas, and Iowa. These are the three DC-hosting states where bulk Secretary of State (SoS) data are freely available. We then ran the same within-site ring DiD with trend-break audit on annual new-LLC formation counts at the fine and coarse rings (`cf_220`–`cf_228`). For Virginia and Texas we geocoded ∼1.4M and ∼890k unique addresses via the Census batch geocoder. Iowa publishes already-geocoded home-office coordinates. We also pulled the Census Business Formation Statistics (BFS) county annual file for a national county-level robustness check.

The result is a sharp contextual divergence:

Table 10: New business formation around hyperscale openings, by state and ring

State	Setting	n HS DCs	1–2 km		2–5 km	
			detrend	p	detrend	p
Virginia	dense exurban (NoVA)†	108	−3.6%	0.023	−3.8%	0.016
Texas	mixed Austin/Dallas	14	+5.2%	0.58	−9.2%	0.31
Iowa	rural Midwest	23	+11.8%	0.043	+10.6%	0.15
National county-level BFS:						
HS host counties	—	38 vs 2,520 controls	+1.4% detrended ($p = 0.49$)			

Notes: VA from Virginia SoS LLC filings 2014–2022, geocoded via Census batch (`cf_221`: 396,937 LLCs, 92% match rate within the 50 km catchment). TX from Texas franchise tax taxpayer addresses 2014–2026, geocoded similarly (`cf_223`: 1.54M filings, 90% match rate). IA from data.iowa.gov active business entity list 2014–2026 with pre-geocoded `POINT(lon lat)` home-office coordinates (`cf_227`: 211k active entities post-2014; the IA dataset is active-only, which is the binding measurement caveat for this row). National county-level row from Census BFS county annual file 2005–2024 (`cf_226`), where "host" is a county with any v8c hyperscale DC and "control" is a non-DC county ¿50 km from any DC. All cells from within-site ring DiD with trend-break specification (same as the matrix outcomes in Table 1). † NoVA = Northern Virginia. *,**,***: $p < 0.10, 0.05, 0.01$.

This is a flow measure of new formations, not exits. The dependent variable is the count of newly registered LLCs and corporations in each ring-month cell. We do *not* observe closures or dissolutions of existing businesses. The Virginia −3.6 to −3.8% in the 1–5 km ring is therefore a reduction in the rate at which would-be operators choose to form a new entity in the immediate commercial neighborhood. It is not evidence that existing businesses are closing. The Iowa +11.8% is, symmetrically, an increase in the formation flow rather than a stock-level expansion of incumbents.

Three plausible mechanisms in the exurban case are consistent with reduced entry: commercial-rent capitalization, parcel-availability constraints from the DC footprint and supplier buildout, and the demand shadow of a small on-site DC workforce. Our identification does not separate them. The result fits directly into the activation-with-redistribution reading in §7. The same hyperscale openings that raise aggregate commercial spending, draw in-migrants (Figure 2), and lift residential rents also suppress the small-entity entry margin in the immediate commercial neighborhood, with the negative effect concentrated where the rent and migration pressures are largest.

The three-state pattern in Table 10 is suggestive of a gradient in the local rent response, but should be read with two qualifications. First, with only three observations, the cross-state ordering is suggestive rather than estimated. A different set of states could overturn the sign. Second, the Iowa cell is the most fragile of the three. The Iowa Secretary of State dataset is an active-only registry, so dissolved entities are absent. The +11.8% at 1–2 km therefore partly reflects survivorship rather than entry alone.

With those qualifications, the three states sort along a plausible rent-pressure gradient. Virginia, where the within-site hyperscale rent lift at 5–10 km is among the largest in the sample (point estimates above +10% in the NoVA cluster), shows the negative formation effect of −3.6 to −3.8%. Iowa, a slack rural rental market with low baseline price-of-entry inferred from state-level supply-elasticity data, shows the positive formation effect of +11.8%. Texas, where the hyperscale footprint is concentrated in the mixed Austin and Dallas exurbs, falls in between and is noisy (+5.2% at 1–2 km, −9.2% at 2–5 km, neither significant).

The sign pattern would be consistent with the land-cost crowd-out mechanism if cost-of-entry differences drive the cross-state ordering. We do not read the three-cell pattern as confirming the mechanism. A more definitive cross-jurisdiction test would require state-of-incorporation panels from additional DC-hosting states (Arizona, Washington, North Carolina) and ideally an active-plus-dissolved registry like Virginia's. We have filed the relevant requests, but the data are not yet in hand.

An entity-type decomposition (`cf_231`) localizes the effect to non-passive-holding entities, consistent with genuine local activity rather than registered-agent shells, though it does not prove it. We classified each filing into one of six buckets ordered by priority: passive-holding (name matches *Holdings* / *Properties* / *Investments* / *Trust* / *Capital* / *Ventures*), foreign (out-of-state registration), nonprofit, professional (PLLC / Professional Corporation), self-managed (registered-agent address equals principal address, a sole-proprietor LLC indicator), and standard-operating (everything else). The contextual divergence is concentrated in real-economy buckets in both states.

Table 11: New business formation by entity type, hyperscale rings

State	Entity-type bucket	n records	1–2 km		2–5 km	
			detrend	p	detrend	p
VA	*Self-managed (sole-prop LLC)*	272k	−3.0%	0.028	−3.2%	0.028
VA	Standard operating	135k	−2.0%	0.13	−1.9%	0.20
VA	Passive holding	29k	−1.0%	0.25	−0.3%	0.70
VA	Foreign	5k	−0.1%	0.76	−1.0%	0.029
IA	*Standard operating*	78k	+8.4%	0.044	+5.4%	0.36
IA	Self-managed	72k	+4.9%	0.19	+1.2%	0.76
IA	Passive holding	20k	+1.4%	0.63	+6.3%	0.28
IA	Foreign	28k	−0.1%	0.73	+2.1%	0.26
IA	Nonprofit	10k	−0.6%	0.82	+0.5%	0.85
IA	Professional	3k	−0.9%	0.46	−0.1%	0.97

Notes: Same ring-DiD + trend-break specification as Table 10, restricted to the indicated entity-type bucket. "Self-managed" = principal address equals registered-agent address (a proxy for sole-prop LLC structure); "Passive holding" = name matches a holding/investment regex (`cf_231`); "Foreign" = registered outside the host state. VA: file is all LLCs, so no nonprofit / professional row; the self-managed indicator is built from the (principal, RA) address comparison. IA: explicit `Corporation Type` field carries the nonprofit / professional / foreign tags; self-managed is again the (principal, RA) match. Buckets are mutually exclusive and applied in the priority order listed in the text. $^{*}, ^{**}, ^{***}$: $p < 0.10, 0.05, 0.01$.

In Virginia, the aggregate −3.8% at 2–5 km is concentrated in *self-managed LLCs* (−3.2%,

$p = 0.028$), sole-proprietor or single-member structures with no separate registered agent. These are the smallest, most personally exposed operators (the local handyman, the single-family-rental LLC, the side-business consulting LLC). They are also the most cost-sensitive to commercial-rent pressure and the most likely to substitute to a cheaper address farther out when rents rise in the immediate ring. The standard-operating bucket is also directionally negative (-1.9%, $p = 0.20$) but the effect is smaller. Passive holdings, foreign entities, nonprofits, and professional-service LLCs are essentially null. The negative entry effect is not a shell-LLC phenomenon.

In Iowa, the aggregate +11.8% at 1–2 km is concentrated in *standard-operating entities* (+8.4%, $p = 0.044$), real businesses with professional structure (separate registered agent, non-shell name). Self-managed and passive-holding categories are directionally positive but not significant. Foreign, nonprofit, and professional entities are null. The positive entry effect is also *not* a shell or passive-investment artifact.

The reduced entry is not just redirected to nearby rings. A natural alternative reading of the VA negative is that would-be operators simply file at a cheaper address slightly farther out, in which case the 5–10 km and 10–25 km rings should show a positive offset. They do not: VA 5–10 km detrended is -1.1% ($p = 0.20$) and 10–25 km is -1.4% ($p = 0.11$), both directionally negative. The reduced entry in the immediate ring appears to be either a net loss of would-be formations from the broader catchment, realized at locations outside the 50 km comparison radius, or not realized at all.

The national county-level BFS row is null (+1.4% detrended, $p = 0.49$), consistent with counties being too coarse to isolate the ring-level signal from surrounding null space. This pattern underscores why fine-grained ring-level data are essential to detect the sign-reversal between rural and exurban contexts.

As a robustness slice, we link each Secretary-of-State filing to the Veridion firm master by normalized name, ZIP, and state, which recovers a NAICS code for about 2 percent of records. This subset is the filings with a detectable digital footprint, web presence, and operational track record by Veridion's coverage standards. On this much-smaller subset, which is selection-biased toward established operating businesses, the contextual effects attenuate substantially. The Virginia -3.8% at 2–5 km becomes -0.6% (NS), and the Iowa +11.8% at 1–2 km becomes +0.5% (NS).

Two interpretations fit. The pessimistic read is that shell-LLC behavior drives the aggregate results without reflecting real economic activity. Our preferred read is that Veridion's universe is selection-biased toward established firms with digital footprints. By construction it excludes the population we want to measure, which is newly-forming entities responding to a local treatment. New small operators have not yet had time to develop a Veridion-visible web presence. And the entity-type decomposition above shows the response is concentrated in self-managed and standard-operating buckets, the categories most under-represented in Veridion relative to their share of SoS filings.

We therefore report the SoS aggregate as the primary measurement and the Veridion-matched subset as a complementary slice. The slice tells us the response runs through smaller, newer, not-yet-digital entities rather than through already-established operating firms.

Two interpretive caveats remain. First, the Iowa data are active-only (dissolved entities are not included), which biases the IA flow count toward formations that survived. The direction of this bias is not signed without observing exits. If DC-adjacent formations survive longer, the active-only panel could amplify the positive estimate. If omitted short-lived rural formations are more common away from DC catchments, the panel could attenuate it instead. The entity-type decomposition mitigates the concern but does not remove it. The positive effect is concentrated in standard-operating entities (the modal real-business form), not in self-managed or passive-holding shells that might inflate active-only counts.

Second, the Texas hyperscale sample is small ($n = 14$) and serves more as a directional check than a power test. North Carolina ($\sim$ 15 HS DCs) and Arizona ($\sim$ 8) would be informative intermediate-density cases. Both require paid or form-mediated access to bulk SoS data and are reserved for the revision.

The firm-entry response to a hyperscale opening is not a structural feature of the data-center treatment itself. It depends on the local density and supplier base of the host area. In rural Iowa, hyperscale arrival is followed by increased *entry* of real operating businesses (standard-operating LLCs and corporations) in the immediate 1–5 km ring. These are the kinds of entities needed to support the build and the in-migrating workforce.

In dense exurban Virginia, the same treatment is followed by reduced *entry* of sole-proprietor and small operating entities in the same ring, with no offsetting increase at nearby rings. This is consistent with rent capitalization, parcel-availability constraints, and a weak local demand response from the small DC operations workforce. The two effects cancel at the national county-level aggregate.

The contextual reading aligns with the rent and capital findings already in the matrix. Rent rises and a capital footprint appears in both contexts. But the flow of new economic actors into the immediate commercial neighborhood responds to those pressures in opposite directions, depending on whether the catchment has spare commercial supply (rural) or not (exurban).

Data Availability Statement

This study combines publicly available data with several proprietary datasets obtained under license. All code that constructs the analysis datasets from the raw sources and reproduces every table and figure will be deposited in the AEA Data and Code Repository. The deposit includes the publicly available inputs and all author-constructed files, namely the data-center registry, the operating and construction-start dates, and the announced-then-cancelled cohort. The proprietary inputs cannot be redistributed. For each, we report the provider and access terms and include the exact product identifiers, date ranges, geographic filters, and extraction code, so that a licensed user can regenerate the facility-by-ring panels and reproduce all results.

Publicly available data.

- Satellite nighttime lights: VIIRS Day/Night Band Monthly Composite (NOAA, product VCMCFG), accessed through Google Earth Engine.
- Daytime imagery for construction dating: Sentinel-2 and Landsat surface reflectance (ESA and USGS), accessed through Google Earth Engine.
- Data-center registry: base layer from the IM3 Open Source Data Center Atlas (Pacific Northwest National Laboratory), augmented by the authors from operator press releases, state economic-development announcements, the Good Jobs First subsidy tracker, trade press, DatacenterMap, Cloudscene, and data-center REIT filings. The compiled registry, dates, and cancelled cohort are in the deposit.
- Construction timing: Federal Aviation Administration Obstruction Evaluation / Airport Airspace Analysis filings (Form 7460-1).
- Employment and establishments: Quarterly Census of Employment and Wages (Bureau of Labor Statistics) and County Business Patterns (Census Bureau).
- Migration and income: IRS Statistics of Income county-to-county migration data.
- Housing supply: Census Building Permits Survey. Population: 2020 Census Centers of Population (tract population-weighted centroids).
- Workplace employment: LEHD Origin-Destination Employment Statistics (LODES, Census Bureau).
- Business formation: Census Business Formation Statistics, and state business-registration bulk files for Virginia, Texas, and Iowa (Secretaries of State and data.iowa.gov).
- Local public finance: Census of Governments and the Annual Survey of School System Finances (F-33).
- Electricity: U.S. Energy Information Administration Form 861.
- Instrument-search inputs: the InterTubes long-haul fiber map [Durairajan et al., 2015]; the 1980 county urban-college share (IPUMS NHGIS); and transmission-substation locations (Homeland Infrastructure Foundation-Level Data).

Proprietary data (available under license, not redistributable).

- Job postings: LinkUp Job Records [LinkUp, 2025] and WageScape salary postings [WageScape, 2022].
- Residential rents: RentHub Rental Data [RentHub, 2022].
- Consumer spending: SafeGraph Spend Patterns [SafeGraph, 2022].
- Foot traffic: Advan Weekly Patterns [Advan Research, 2025].

- Firmographics: Veridion firm master data.

The five Dewey-distributed datasets are available to researchers through a subscription to Dewey Data (https://www.deweydata.io); Veridion firmographics are available under license from Veridion (https://veridion.com). The authors accessed these data under institutional license and are not permitted to redistribute the raw files. The deposit provides the dataset identifiers and the extraction and aggregation code required to rebuild the analysis files from a licensed copy.

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